

Lab 1: Requirements Engineering & UML Use-Case Modelling

Incident Escalation & On-Call Rotation Engine

Deliverable 1 — Use-Case Flow Specification

Use Case ID: UC-01

Use Case Name: Handle Incident Alert

Primary Actors

- Monitoring System
- On-Call Engineer

Supporting Actor

- Incident Commander

Preconditions

1. The on-call rotation schedule has been configured with a valid primary and secondary engineer.
2. The Monitoring System is connected to the Incident Escalation & On-Call Rotation Engine.
3. At least one notification channel such as SMS, email, or webhook is configured.
4. The assigned on-call engineer has an active account in the system.

Trigger

The Monitoring System generates and sends a new incident alert to the Incident Escalation & On-Call Rotation Engine.

Main Success Scenario

1. The Monitoring System sends an incident alert to the system.
2. The system receives the alert and records its incident ID, severity, timestamp, and incident details.
3. The system identifies the active primary on-call engineer from the current rotation schedule.
4. The system dispatches the incident notification through the configured notification channels.
5. The on-call engineer receives the incident notification.

6. The on-call engineer reviews the incident details.
7. The on-call engineer acknowledges the incident within 5 minutes.
8. The system records the acknowledgment time and the identity of the acknowledging engineer.
9. The system changes the incident status to Acknowledged.
10. The system stops further automatic escalation for the incident.
11. The use case ends successfully.

Postconditions

1. The incident is stored in the system with an Acknowledged status.
2. The responsible on-call engineer is recorded against the incident.
3. The acknowledgment timestamp is stored.
4. No further escalation occurs unless another escalation condition is triggered separately.

Alternate Flow — A1: Incident Not Acknowledged Within 5 Minutes

- A1.1 The primary on-call engineer does not acknowledge the incident within 5 minutes.
- A1.2 The system detects that the acknowledgment timeout has expired.
- A1.3 The system identifies the secondary engineer from the configured on-call rotation.
- A1.4 The system escalates the incident to the secondary engineer.
- A1.5 The system sends the incident notification to the secondary engineer through the configured notification channels.
- A1.6 The system records the escalation event, timestamp, and escalation level.
- A1.7 The Incident Commander may be notified according to the escalation configuration.
- A1.8 The incident remains open until an assigned responder acknowledges it.